Lesson notes · for you

# Careers in cyber security

Cyber security · Level 4 · Lesson 4

| Time | Materials |
| --- | --- |
| 17 min | A phone or computer for the lab; Paper for the job card |

## Learning goals

- Describe several jobs in cyber security.
- Explain why permission comes first, and what to do if you find a weakness.
- Plan your own first steps.

## Activity: Defender for a day

1. Warm-up (3 min): Read four short job descriptions and guess each job's name.
2. Lab (6 min): In the "Teach the machine" lab, teach the filter with safe and phishing examples. Write one tricky message it gets wrong, and say what example would fix it.
3. Sort (5 min): Sort six situation cards into "allowed" and "not allowed", and say why: a CTF for students; changing your own phone's settings; logging in to a classmate's account "to test it"; testing the campus network "to help" without asking; reporting a weakness privately; a club practice lab.
4. Plan (3 min): Write one step you will take this term.

**Prepare six situation cards and four job descriptions on paper, one set per group.**

## Discussion

- Why is "just looking" without permission still a crime?
- What should you do if you notice a weakness by accident?
- Which of your university modules would help most in security, and why?

## Check yourself

- One security job, and one first step I could take.

**Try it this week**
In the lab, train a phishing filter with examples, then find a message it gets wrong. Then pick one security job, find out what a day in it looks like, and tell a friend.

Worksheet

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Name _____ Class _____ Date _____

**1** **While checking your own results, you see another student's details on the university's student portal by mistake. What's the ethical next step?**

- ◯ Tell the university's IT team privately, and go no further
- ◯ Look around to see what else shows up
- ◯ Post a warning on social media

**2** **Match each job to what it does.**

| | | |
|---|---|---|
| Security analyst | • • • • | Tests systems, with written permission |
| Forensic investigator | • • • • | Makes sure people's data is treated fairly |
| Ethical hacker | • • • • | Watches alarms and checks reported messages |
| Privacy officer | • • • • | Works out what happened after an attack |

**3** **Looking around someone else's system without permission is a crime in almost every country, even if you change nothing.**

- ◯ True
- ◯ False

**4** **Which are good first steps towards a security career for a student?**

- ☐ Test your university's website to show off your skills
- ☐ Join or start a computing club
- ☐ Try classmates' passwords to show them they're weak
- ☐ Take free courses and practise on labs built for practice
- ☐ Enter capture-the-flag competitions made for students

✎ **Which security job sounds most like you, and why? What is one thing you could do this term to find out more?**

Check your answers

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**1** **Tell the university's IT team privately, and go no further**

Reporting privately lets the people who can fix it act before anyone is harmed.

**2** **Security analyst — Watches alarms and checks reported messages**
**Forensic investigator — Works out what happened after an attack**
**Ethical hacker — Tests systems, with written permission**
**Privacy officer — Makes sure people's data is treated fairly**

Security is teamwork: some people watch, some investigate, some test with permission and some protect people's rights.

**3** **True**

In almost every country, getting in without permission is the crime, whether or not you change anything.

**4** **Join or start a computing club; Enter capture-the-flag competitions made for students; Take free courses and practise on labs built for practice**

Clubs, student competitions and practice labs build real skills legally.